

2024 Ph H2 Q15

Section: Electricity

Topic: Semiconductors & Solar Cells

Summary of Question

A set of garden lights is powered by an array of solar cells. The set has 10 LEDs connected in parallel, each in series with a resistor of combined resistance $220\ \Omega$.

Tasks: (a) Calculate the total resistance. (b) A solar cell is a p-n junction. (i) Name the effect. (ii) Explain how a potential difference is produced.

(a) Total resistance

For N equal resistors in parallel, $R_{\text{total}} = R/N$.

Here, $R_{\text{total}} = 220 / 10 = 22\ \Omega$.

Final answer: $22\ \Omega$

(b)(i) Name of effect

Answer: the photovoltaic effect.

(b)(ii) Explanation using band theory

- Photons of light with sufficient energy excite electrons from the valence band to the conduction band in the semiconductor.
- This creates electron-hole pairs.
- The built-in electric field at the p-n junction drives electrons towards the n-side and holes towards the p-side.
- This separation of charge carriers produces a potential difference across the junction.

Revision Tips

For resistors in parallel: $1/R_{\text{total}} = 1/R_1 + 1/R_2 + \dots$ For identical resistors, $R_{\text{total}} = R/N$.

The photovoltaic effect is the basis of solar cells (not to be confused with the photoelectric effect on metals).

Band theory: photons create electron-hole pairs; the junction field separates them, giving a measurable voltage.